Mathematical Skills for Computing

11th November 2024

Examination Paper

Answer ALL questions.

Clearly cross out surplus answers.

Time: 3 hours

The maximum mark for this paper is 100.

Any reference material brought into the examination room must be handed to the invigilator before the start of the examination.

The following guidelines are to be adhered to in the use of non-programmable calculator in the examination:

- **The calculator must be silent, with a visual display only.**
- **Candidates are not allowed to share calculators during the examination.**

SECTION A – Multiple choice questions
Circle ONE (1) correct answer from A, B, C or D for each question.
Each question is worth 1 mark.

Marks

Question 1

Simplify $\frac{a^6}{a^4} \times a^{-3}$ **1**

A: a **B**: a^{-1}

C: a^7 **D**: a^{13}

Question 2

Simplify the expression **1**
$(b^c)^d$

A: b^{c+d} **B**: b^{c-d}

C: b^{cd} **D**: $b^{\frac{c}{d}}$

Question 3

Simplify the following expression **1**
$7e^2 + 2e^3 + (5e \times 3e)$

A: $2e^3 + 22e^2$ **B**: $7e^2 + 2e^3 + 15e^2$
C: $9e^5 + 15e^2$ **D**: $24e^7$

Question 4

If D = {1, 4, 10, 12, 13} and E = {4, 10} find $D \cup E$ **1**

A: {1, 4, 10, 12, 13) **B**: {1, 4, 4, 10, 10, 12, 13}
C: {1, 12, 13} **D**: {7}

Question 5

If D = {1, 4, 10, 12, 13} and E = {4, 10} find $D - E$ **1**

A: {3) **B**: {1, 13}
C: {1, 12, 13} **D**: {26}

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Marks

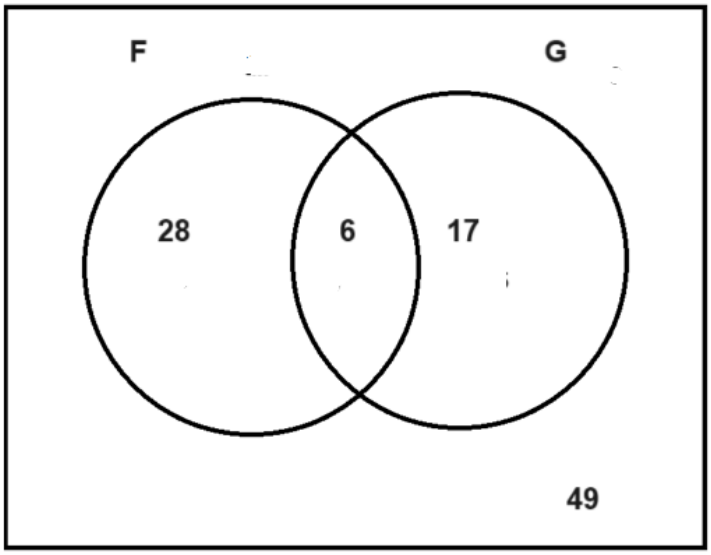
Question 6

If F = {1, 2, 5, 19, 23, 30} find n(F) **1**

A: 6
B: 30
C: {1, 30}
D: 1

Question 7

A group of European travellers are surveyed about the languages they speak. The responses are displayed in the Venn diagram below where F represents French and G represents German. **1**



Select the option that gives the number of European travellers who took part in the survey.

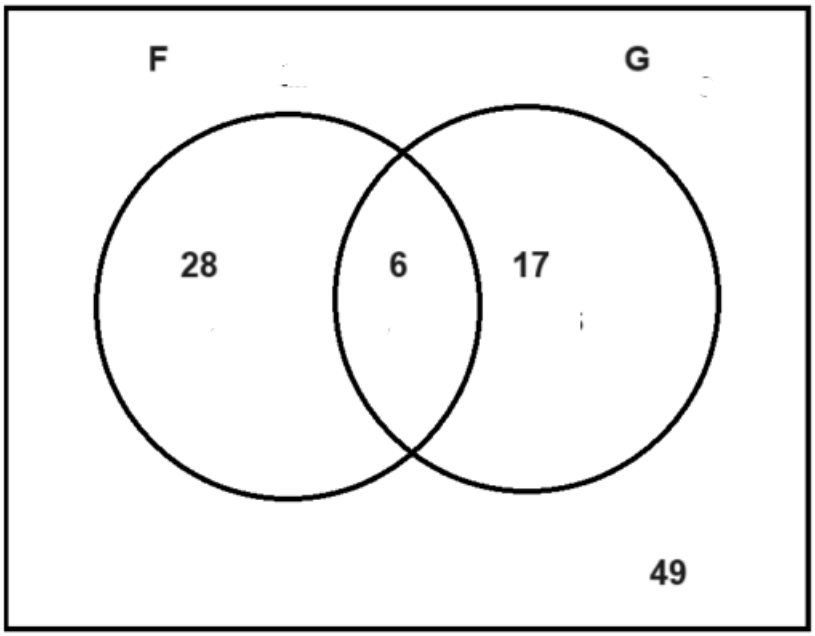
A: 49
B: 100
C: 51
D: 45

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Marks

Question 8

Using the same Venn diagram as in Question 7. **1**



Select the option that gives the number of European travellers who do not speak French or German.

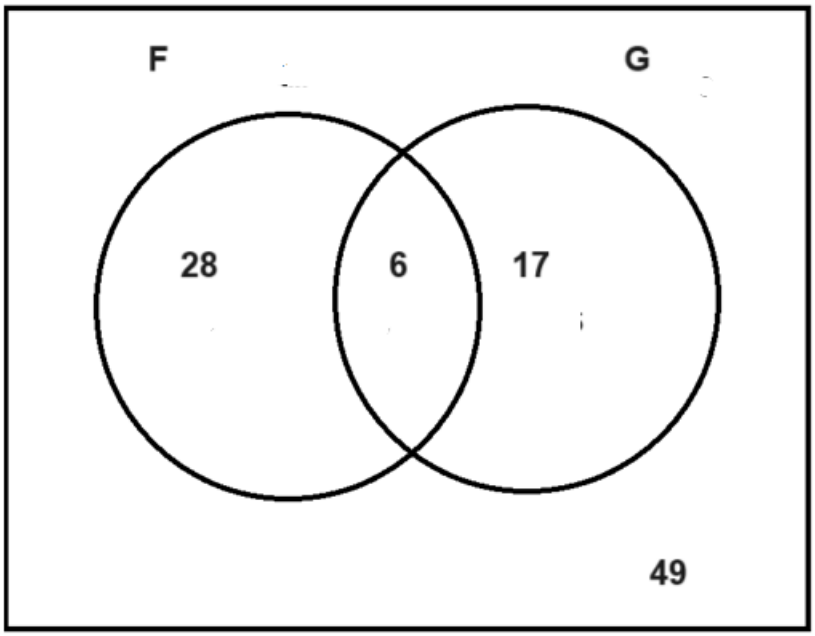
A: 49
B: 100
C: 51
D: 45

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Marks

Question 9

Using the same Venn diagram as in Question 7. **1**



Select the option that gives the number of European travellers who speak both French and German.

A: 49
B: 23
C: 34
D: 6

Question 10

The matrix R is given as **1**

$$\begin{pmatrix} 5 & 3 & 7 \\ -2 & \frac{3}{8} & 0 \\ 2 & 4 & 1 \end{pmatrix}$$

Find RI

A: $\begin{pmatrix} 5 & 3 & 7 \\ -2 & \frac{3}{8} & 0 \\ 2 & 4 & 1 \end{pmatrix}$

B: $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

C: $\begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$

D: $\begin{pmatrix} -5 & -3 & -7 \\ 2 & -\frac{3}{8} & 0 \\ -2 & -4 & -1 \end{pmatrix}$

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Question 11

Find the determinant of Q if Q = $\begin{pmatrix} 4 & -3 \\ 2 & 3 \end{pmatrix}$ **1**

A: -18
B: 6
C: 18
D: -17

Question 12

TWO (2) matrices are given as **1**
$M = \begin{pmatrix} 2 & 5 \\ -1 & 3 \end{pmatrix}$ and $N = \begin{pmatrix} 1 & 2 \\ 4 & 5 \end{pmatrix}$

Calculate $M - N$

A: $\begin{pmatrix} -1 & -3 \\ 5 & 2 \end{pmatrix}$
B: $\begin{pmatrix} 2 & 10 \\ -4 & 15 \end{pmatrix}$
C: $\begin{pmatrix} 3 & 7 \\ 3 & 8 \end{pmatrix}$
D: $\begin{pmatrix} 1 & 3 \\ -5 & -2 \end{pmatrix}$

Question 13

Name the type of matrix shown here **1**

$$\begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

A: Scalar matrix
B: Triangular matrix
C: Identity matrix
D: Null matrix

Question 14

Select the option that gives the order of the following matrix. **1**

$$\begin{pmatrix} 1 & 0 \\ 0 & 2 \\ 3 & 4 \\ 8 & 2 \end{pmatrix}$$

A: 2 × 4
B: 4 × 2
C: 8
D: 2

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Question 15

Select the option that is an example of secondary discrete data. **1**

A: Census data on number of households

B: Research article on heights of women

C: Survey on amount of time people spend on laptop

D: Survey on number of hours people work each work

Question 16

A data set is given by M ={125, 125, 151, 132, 132, 141, 120, 125, 115, 141 } **1**
Calculate the mode of the data set.

A: 111

B:132

C:125

D: 30

Question 17

A data set is given by M ={125, 125, 151, 132, 132, 141, 120, 125, 115, 141 } **1**
Calculate the mean of the data set.

A: 130.7

B: 125

C:111

D: 30

Question 18

A data set is given by M ={125, 125, 151, 132, 132, 141, 120, 125, 115, 141 } **1**
Calculate the median of the data set.

A: 128.5

B: 125

C:130.7

D: 30

Question 19

EIGHTY (80) people were surveyed and asked about the method of transport they used to travel to work. FIFTEEN (15) people said that they cycled. **1**
Select the angle on a pie chart which would be used the represent this.

A:15

B: $\frac{3}{16}$

C: 67.5

D: 65

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Marks

Question 20

Using the same data as in Question 19, a sector with an angle of 144° represents the number of people who travel to work by train. Calculate the number of people who travel to work by train. **1**

A: 32

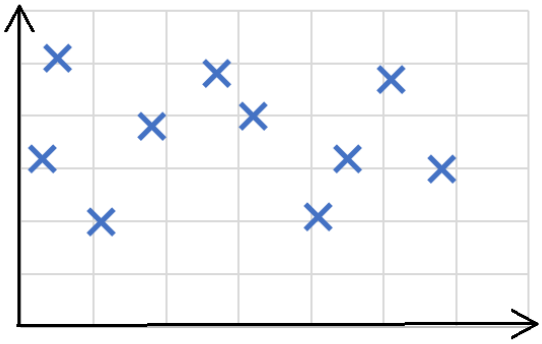
B: 144

C: $\frac{2}{5}$

D: 48

Question 21

Select the option which best describes the type of correlation shown in the scatter graph below. **1**



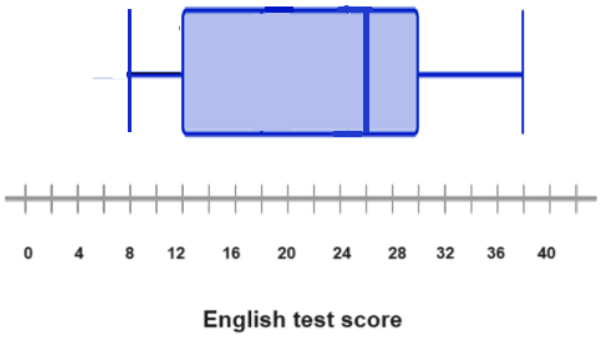
A: Strong positive correlation.
B: Weak positive correlation
C: No correlation
D: Weak negative correlation

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Question 22

The box plot below displays the scores achieved by a group of students in an English test. **1**

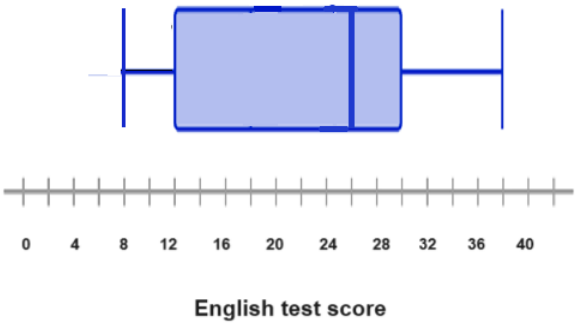


Find the interquartile range.

A: 26
B: 30
C: 12
D: 18

Question 23

The box plot below displays the scores achieved by a group of students in an English test. **1**



Find the range.

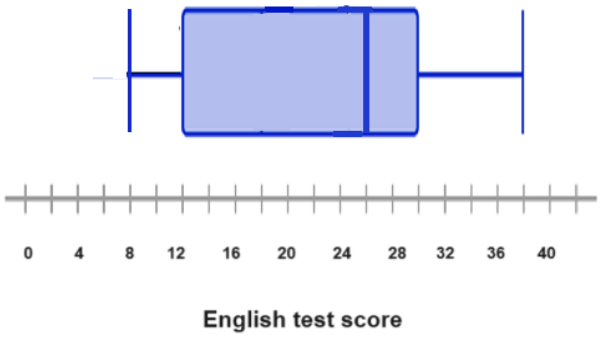
A: 30
B: 18
C: 8
D: 38

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Marks

Question 24

The box plot below displays the scores achieved by a group of students in an English test. **1**



Find the median score.

A: 26
B: 12
C: 18
D: 30

Question 25

The variance of a set of data is 90.25. Find the standard deviation. **1**

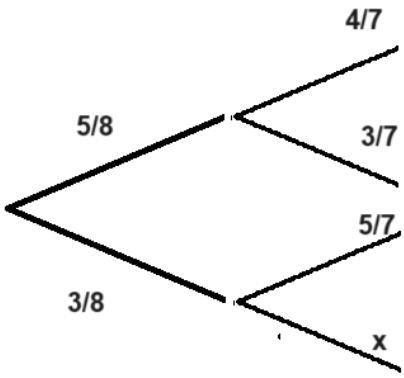
A: 180.5
B: 8145.0625
C: 9.5
D: 45.125

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Marks

Question 26

Find the value of x in the tree diagram below: **1**



A: $\frac{3}{7}$

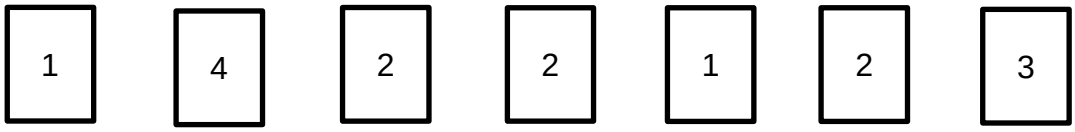
B: $\frac{2}{7}$

C: $\frac{2}{8}$

D: $\frac{3}{8}$

Question 27

Here are some number cards. **1**



ONE (1) card is selected at random. Calculate the probability that it will have a number 2 on it.

A: $\frac{2}{7}$

B: $\frac{3}{7}$

C: $\frac{5}{7}$

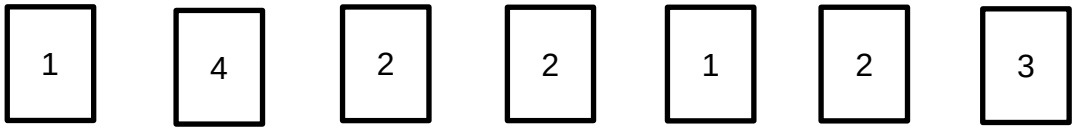
D: $\frac{7}{3}$

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Marks

Question 28

Here are some number cards. **1**



ONE (1) card is selected at random. Calculate the probability that it will **NOT** have a number 1 on it.

A: $\frac{5}{7}$

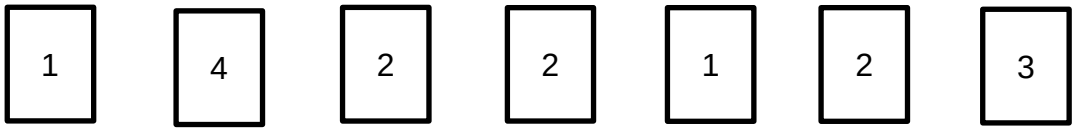
B: $\frac{2}{7}$

C: $\frac{6}{7}$

D: $\frac{1}{7}$

Question 29

Here are some number cards. **1**



TWO (2) cards are selected at random without replacement. Calculate the probability that they will **both** have a number 2 on them.

A: $\frac{3}{7}$

B: $\frac{2}{7}$

C: $\frac{5}{7}$

D: $\frac{1}{7}$

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Marks

Question 30

A meal deal includes a drink, a sandwich and a bag of crisps. **1**
There are EIGHT (8) drinks, SIX (6) sandwiches, and FIVE (5) flavours of bags of crisps to choose from. Work out the total number of ways of choosing a drink, a sandwich and a bag of crisps.

A: 8
B: 5
C: 19
D: 240

Total 30 Marks

Questions continue on the next page

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SECTION B
Answer ALL questions. You must show your workings.

Question 31

a) Simplify the following:

i) $\frac{4x}{5} \div \frac{3x^2}{7}$ **2**

ii) $\frac{x^2 - 49}{x - 7}$ **2**

b) Transpose the following formula to make b the subject: **2**

$$a = \frac{5d}{b^2 + c}$$

c) Solve the following equation and find the value of f: **2**

$$5(3f - 2) = -2.5$$

d) Solve the following quadratic equation by using the Quadratic Formula: **2**

$3x^2 - 7x + 1 = 0$

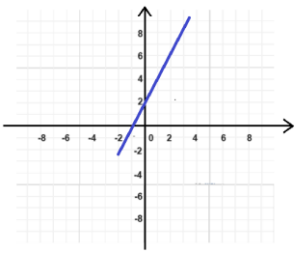
Leave your answer in surd form.

Total 10 Marks

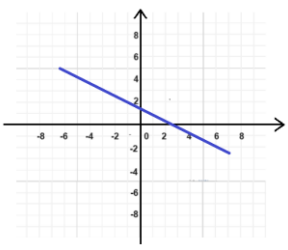
Question 32

a) State which of the following graphs have the steepest negative gradient. **1**

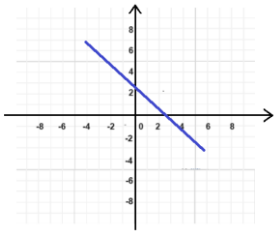
Line 1



Line 2



Line 3



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Marks

b) Consider the pair of simultaneous equations:
$y = 3x - \frac{1}{2}$ and $y = 4x - 1$

i) Construct a table of integer values for each equation for the range. **4**
$-1 \leq x \leq 3$

ii) Draw accurate graphs for both equations on ONE (1) set of axes. Use the graph paper provided. **4**

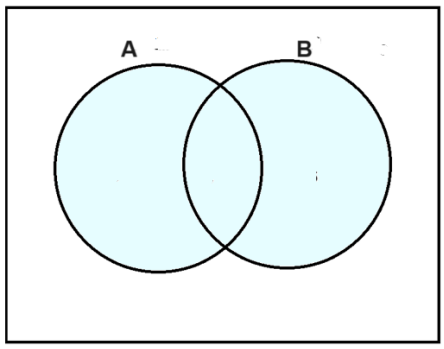
iii) Use the graphs constructed in (ii) to find the solution of the pair of simultaneous equations. **1**

Total 10 Marks

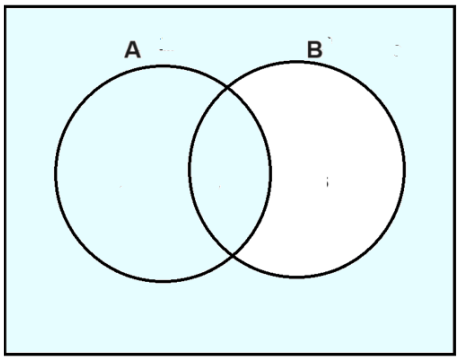
Question 33

a) State the set notation for the shaded areas of each of the following Venn diagrams.

i) **1**



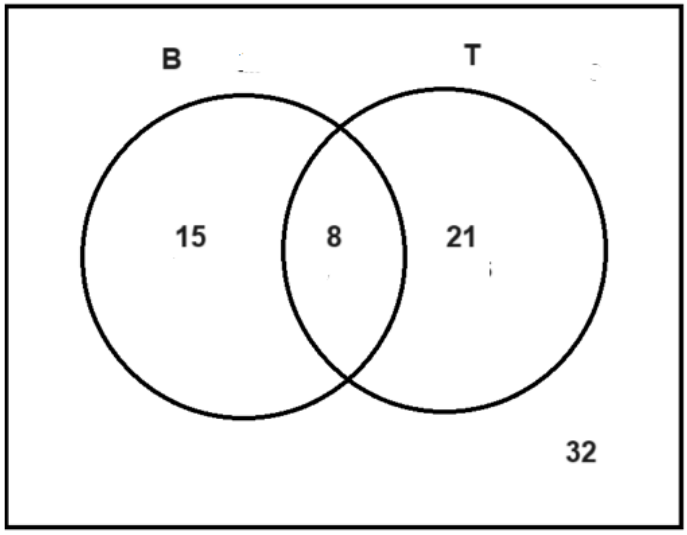
ii) **1**



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Marks

b) The Venn diagram below shows how a group of people travel to work.
B is the event that they travel to work by bus.
T is the event that they travel to work by train.



i) Find $n\,(B \cup T)$ **and** explain what this means. **2**

ii) Find $n\,(B' \cap T)$ **and** explain what this means. **2**

iii) Find $n\,(B' \cap T')$ **and** explain what this means. **2**

c) $A = \{1, 2, 7\}$and $B = \{x: x \in B, x > 4\}$ **2**
State whether A and B are disjoint sets. Explain your answer.

Total 10 Marks

Question 34

a) TWO (2) matrices are given as

$C = \begin{pmatrix} 5 & 2 \\ 1 & 3 \end{pmatrix}$ and $D = \begin{pmatrix} -1 & -2 \\ 4 & 2 \end{pmatrix}$

i) Calculate $C + D$. **1**

ii) Matrix E is represented by 5C. Write down matrix E. **1**

iii) State the transpose of matrix E. **1**

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Marks

b) TWO (2) matrices are given as

$F = \begin{pmatrix} -2 & 1 \\ 3 & 4 \end{pmatrix}$ and $G = \begin{pmatrix} 2 & 3 \\ \frac{1}{2} & 1 \end{pmatrix}$

i) Calculate FG **2**

ii) Using your answer to part (i) show that $FG \neq GF$ **3**

c) Find the matrix that represents the linear transformation. **2**

$H: \begin{pmatrix} x \\ y \end{pmatrix} \rightarrow \begin{pmatrix} 2x \\ 5x - 7y \end{pmatrix}$

Total 10 Marks

Question 35

A group of 122 commuters were surveyed to find the distances they travel to work. The table shows the results.

Distance d (km)	Frequency
$0 \leq d < 6$	6
$6 \leq d < 10$	12
$10 \leq d < 20$	40
$20 \leq d < 40$	48
$40 \leq d < 45$	10
$45 \leq d < 60$	6

a) Complete the following table of frequency densities. **2**

Distance d (km)	Frequency	Frequency density
$0 \leq d < 6$	6	
$6 \leq d < 10$	12	
$10 \leq d < 20$	40	
$20 \leq d < 40$	48	
$40 \leq d < 45$	10	
$45 \leq d < 60$	6	

b) Construct a histogram of the data using your answer to part (a). You should use the graph paper provided to draw your histogram. **6**

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Marks

c) State the modal class. **1**

d) State the median class. **1**

Total 10 Marks

Question 36

The amount of time a group of customers spend in a supermarket is recorded in the table below.

Time, t, (minutes)	Frequency
$0 \leq t < 10$	10
$10 \leq t < 20$	25
$20 \leq t < 30$	51
$30 \leq t < 40$	38
$40 \leq t < 50$	24
$50 \leq t < 60$	12

a) Calculate the cumulative frequency of the data. **1**

b) State within which class interval the lower quartile (Q1) value will lie **and** then calculate an estimate of the value using the formula for the lower quartile. **3**

c) State within which class interval the upper quartile (Q3) value will lie **and** then calculate an estimate of the value using the formula for the upper quartile. **3**

d) Calculate the interquartile range. **1**

e) Calculate the range. **1**

f) Give ONE (1) advantage of using the interquartile range rather than the range to describe the spread of a data set. **1**

Total 10 Marks

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Marks

Question 37

In an experiment, a FIVE (5) sided spinner, numbered ONE (1) to FIVE (5), is spun FOUR HUNDRED (400) times and the results are recorded in the table below.

Score	1	2	3	4	5
Frequency	102	128	41	57	

a) Calculate the relative frequency of the spinner landing on a FIVE (5). Give your answer as a fraction in its simplest form. **2**

b) What is the relative frequency of the spinner landing on a ONE (1)? Give your answer as a fraction in its simplest form. **1**

c) What is the relative frequency of the spinner landing on a ONE (1) or a FOUR (4)? Give your answer as a fraction. **2**

d) What is the theoretical probability of a fair, FIVE (5) sided spinner landing on a ONE (1)?
How many times would you expect a fair, FIVE (5) sided spinner to land on a ONE (1) in a trial of FOUR HUNDRED (400) spins? **2**

e) State whether or not you think the spinner used in the experiment is fair. Give a reason for your answer. **2**

f) State whether the events of the spinner landing on a ONE (1) and on a FOUR (4) are mutually exclusive or not. **1**

Total 10 Marks

End of Paper

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